

Systematic Position and Proper Motion Corrections for Historical Astrometric Catalogs Based on Gaia DR3

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摘要

- (1) Background on systematic errors in historical optical observations of minor bodies;
- (2) Purpose of position and proper motion corrections based on Gaia DR3 unified reference frame;
- (3) Data scope including Gaia series and 15 historical astrometric catalogs;
- (4) Method overview of sky-region-based catalog comparison and systematic bias estimation;
- (5) Main results including debiasing correction tables and their value for minor body orbit determination. Word limit: 250 words.]

Keywords: Astrometry — Astronomical catalogs — Minor planets — Gaia

1. INTRODUCTION

小天体的精确轨道确定依赖于高质量的光学天体测量数据。Gaia 任务 ([Astropy Collaboration et al. 2022](#)) 为天体测量提供了前所未有的精度。历史观测中使用了多种参考星表，这些星表之间存在系统性差异。

早期的星表去偏工作基于 2MASS 等星表进行区域修正。随着 Gaia 数据的发布 ([Astropy Collaboration et al. 2018](#))，星表去偏工作进入了新的阶段。

本工作利用 Gaia DR3 的最新精度优势，对多套历史星表进行系统性的位置与自行偏差修正，为小天体轨道确定提供可靠的数据基础。

2. REFERENCE AND ASTROMETRIC CATALOGS

2.1. Gaia Catalog Series

2013 年以来，Gaia 任务对全天空开展了持续的高精度天体测量观测，并已获得约 18 亿个天体源的位置和光度信息。Gaia 第二次数据发布 (Gaia DR2) 首次向公众提供了大规模包含位置、视差和自行在内的五参数天体测量解，使 Gaia 星表成为研究恒星运动、小天体天体测量以及外部星表系统误差的重要参考数据源。Eggl 等人 (2020) 基于 Gaia DR2 的五参数天体测量结果，构建了区域性星表修正模型，降低了历史星表在小天体测量中的系统偏差。

随后发布的 Gaia EDR3 与 Gaia DR3 在天体测量参数化形式上与 Gaia DR2 保持一致，但在观测时间基线、姿态重建、色差标定以及仪器模型等方面均进行了系统性改进。Gaia DR3 的天体测量解基于约 34 个月的观测数据，相较 Gaia DR2 的约 22 个月显著延长。官方验证结果表明 (引用)，与 Gaia DR2 相比，Gaia DR3 的视差形式不确定度典型降低约 20%，自行形式不确定度降低约一倍以上，反映出天体测量精度的实质性提升。

表 1. Comparison of Gaia DR2 and Gaia DR3 astrometric properties

Property	Gaia DR2	Gaia DR3
Observation time span	~22 months	~34 months
Five-parameter solutions	$\sim 1.3 \times 10^9$	$\sim 5.9 \times 10^8$
Six-parameter solutions	Not available	$\sim 8.8 \times 10^8$
Proper-motion uncertainty	Reference level	~factor 2 improvement
Error distribution	Quasi-Gaussian	Quasi-Gaussian (5p > 6p)
Uncertainty underestimation	Present	Present, better quantified
Systematic noise floor (5p)	Not quantified	$\sim 15 \mu\text{as}$

Gaia DR3 对五参数天体测量解的准入条件进行了更为严格的质量控制。大量在 Gaia DR2 中获得五参数解的源，在 Gaia DR3 中被重新分类为六参数解或仅位置解。验证结果显示（引用），天体测量不确定度在统计上仍近似服从高斯分布，其中五参数解的正态性明显优于六参数解，后者包含了较高比例的非单星系统。

鉴于星表修正方法对参考星表误差统计特性的高度敏感性，本文延续 Eggl 等人（2020）的基本思路，选用 Gaia DR3 中质量受控的五参数天体测量解作为参考星表基础。

2.2. Historical Astrometric Catalogs

不同历史时期的光学观测依赖于多种恒星参考目录。这些目录在观测历元、覆盖范围、源密度和天文测量内容上有所不同，反映了地面和空间天文测量发展的连续阶段。下面对所涉及的星表作简要概述。

Astrographic Catalogue

The Astrographic Catalogue (AC) is one of the earliest large-scale photographic astrometric catalogs, produced within the framework of the international *Carte du Ciel* project initiated at the end of the 19th century. The project employed standardized astrographs to photograph the sky in declination zones assigned to different observatories, with the bulk of the observations carried out between the late 1890s and the first half of the 20th century.

The AC comprises several million stellar positions derived from photographic plates, with typical positional accuracies at the few hundred milliarcsecond level. Owing to its very early mean epoch, AC primarily represents the reference framework of early photographic astrometry and provides the longest available astrometric time baseline among the catalogs considered in this work.

Astronomische Gesellschaft Katalog 1

AGK1 is a fundamental star catalog based on visual meridian-circle observations, covering the northern sky down to approximately 2 degrees in declination. Its observations were obtained mainly between the late 19th century and the early 20th century.

With a limiting magnitude of about 9 and positional accuracies at the arcsecond level, AGK1 represents the reference framework of pre-photographic astrometry. It served as a primary reference catalog for several early photographic surveys, including parts of the Astrographic Catalogue.

Astronomische Gesellschaft Katalog 3

AGK3 is a photographic catalog covering the northern sky, developed as the third installment of the AGK series. Its observations were carried out around the late 1950s, and it introduced proper motion information by re-observing earlier AGK stars.

With typical positional accuracies of about 0.1 arcseconds and proper motion accuracies at the level of several milliarcseconds per year, AGK3 became a widely adopted reference catalog for ground-based astrometry during the 1970s–1990s. The catalog is strictly tied to the FK4 reference frame.

Fourth Fundamental Catalogue

The Fourth Fundamental Catalogue (FK4) defined the B1950.0 optical reference frame and constituted the backbone of mid-20th-century astrometry. Although it contains only 1,535 bright fundamental stars, FK4 provided a highly consistent realization of the reference system at the time of its construction.

Many subsequent ground-based catalogs, including AGK3 and SAO, were directly or indirectly tied to FK4. As a result, its reference frame properties propagated into a large fraction of historical astrometric observations.

Smithsonian Astrophysical Observatory Catalog

The SAO Catalog is a compiled all-sky stellar catalog produced to supply sufficient reference star density for practical astrometric applications. It contains approximately 259,000 stars and was widely used in the early computer era for optical observation reduction, including satellite tracking.

The northern portion of SAO primarily relies on AGK-based data, while the southern sky incorporates data from the Yale and Cape catalogs. This heterogeneous construction reflects the state of global astrometric resources available at the time.

Astrographic Catalog Reference Stars

The Astrographic Catalog Reference Stars (ACRS) catalog was compiled by the U.S. Naval Observatory to provide reference stars for the re-reduction of the Astrographic Catalogue. It combines northern hemisphere data from AGK3 with southern hemisphere data from the Second Cape Photographic Catalogue (CPC2).

ACRS was transformed onto the FK5 reference frame and represents an intermediate step between early photographic catalogs and later compiled all-sky catalogs with improved reference frame consistency.

ACT Reference Catalog

The ACT catalog was constructed by combining early epoch positions from the Astrographic Catalogue with later observations from Tycho-1. This approach significantly improved proper motion accuracy compared to Tycho-1 alone, reaching typical values of a few milliarcseconds per year.

ACT played an important role during the transition period between classical ground-based astrometry and space-based catalogs in the Hipparcos era.

Positions and Proper Motions Catalog

The PPM catalog was developed as a successor to SAO, with the objective of providing a uniform realization of the J2000.0 reference frame. It contains nearly 380,000 stars and incorporates data from AGK3 in the north and FOKAT-S in the south, supplemented by early photographic observations to extend the time baseline for proper motion determination.

Before the advent of CCD-based surveys, PPM represented one of the most advanced compiled ground-based astrometric catalogs.

Yale Zone Catalogs

The Yale Zone Catalogs consist of a series of regional photographic catalogs compiled by the Yale University Observatory. Their observations were mainly conducted between the 1930s and 1950s and focused on equatorial and southern sky regions.

These catalogs were designed to improve the efficiency of stellar position measurements over wide fields and later contributed substantially to several compiled all-sky catalogs. In terms of astrometric precision, the Yale Zone Catalogs typically reach sub-arcsecond accuracy, characteristic of mid-20th-century photographic astrometry.

Hipparcos Catalog

The Hipparcos catalog is the result of the first space astrometry mission and provides a high-precision, nearly rigid optical reference frame. It contains about 118,000 stars with typical positional and proper motion accuracies at the milliarcsecond level, with a mean observational epoch around 1991.25.

Hipparcos established the foundation for all subsequent modern astrometric catalogs and serves as a fundamental link between ground-based and space-based reference systems.

Tycho-2 Catalog

The Tycho-2 catalog combines Hipparcos star mapper observations with data from 144 ground-based catalogs to produce positions and proper motions for approximately 2.5 million stars. Typical positional accuracies are of order 60 mas, with proper motion accuracies around 2–3 mas/yr.

Prior to Gaia, Tycho-2 was the most widely used dense all-sky catalog for high-precision astrometric applications.

USNO-A2.0 Catalog

USNO-A2.0 is a very large astrometric catalog generated from scans of photographic Schmidt plates, containing over 500 million sources. The catalog does not provide proper motion information; all positions correspond to the plate observation epochs, typically around 1950 for POSS-I and around 1980 for POSS-II and SRC plates.

Due to its extremely high source density, USNO-A2.0 was extensively used for optical observation reduction before the availability of dense CCD-based catalogs.

Guide Star Catalog 1.2

GSC 1.2 is a re-calibrated version of the original Guide Star Catalog, developed primarily to support space telescope pointing requirements. It contains roughly 19 million stars and is based on photographic plate material, with positional accuracies at the few tenths of an arcsecond level.

Although it does not include proper motions, GSC 1.2 has been widely used as an auxiliary reference catalog for ground-based astrometry.

UCAC 2

UCAC2 is the second release of the U.S. Naval Observatory CCD Astrograph Catalog and represents the transition from photographic plates to CCD-based astrometry. Its observations were conducted between approximately 1998 and 2003 and cover most of the sky from -90° to about $+40^\circ$ in declination.

The catalog contains about 48 million stars, with positional accuracies ranging from roughly 15 to 70 mas and proper motion accuracies at the level of 1–7 mas/yr.

UCAC 4

UCAC4 is the final release of the UCAC project and provides all-sky coverage. It includes over 113 million stars and incorporates improved proper motion determinations through the use of additional early-epoch photographic material.

With typical positional accuracies of about 15–20 mas for stars in the magnitude range 10–14, UCAC4 represents the mature stage of ground-based CCD astrometric catalogs prior to the Gaia era.

3. STAR POSITION AND PROPER MOTION CORRECTIONS

3.1. Basic Framework of the Correction Method

本研究的方法总体上继承了 Chesley 等人 (2010)、Farnocchia 等人 (2015) 以及 Eggl 等人 (2020) 所提出的星表系统误差去偏框架。通过将全天空划分为等面积天区, 在局部天区尺度上比较测试星表与参考星表中恒星的位置与自行差异, 从而估计星表的系统性偏差。基于 Gaia 星表在天体测量精度方面的显著优势, 本研究选用 Gaia DR3 作为统一的参考星表。

在实现过程中, 不同星表之间首先被统一到一致的比较基准。参考星表与测试星表的恒星位置与自行均被转换至 J2000 历元, 对于早期星表同时考虑参考系的统一。随后, 采用 HEALPix 等面积像素化方案将全天空划分为 49,152 个天区, 对应单个天区面积约为 0.8 deg^2 , 并在每个天区内对测试星表与 Gaia DR3 中的恒星源进行空间交叉匹配。

恒星匹配在参考位置附近设定有限的搜索半径内完成 (本文采用 $1''$ 的搜索范围), 以识别潜在的匹配候选源。当存在多个候选匹配时, 采用最近邻匹配策略, 并结合 Farnocchia 等人 (2015) 提出的异常值剔除方法, 仅保留角距离显著优于其他候选者的唯一匹配。同时施加一一对应约束, 以避免单个 Gaia 恒星源与多个测试星表源发生关联, 从而降低密集星场中伪匹配和多对一匹配对系统偏差统计的影响。

在完成恒星交叉匹配后, 对每个天区内匹配恒星在位置与自行上的差异进行统计分析。所有差异均在 J2000.0 历元下计算, 分别对应赤经与赤纬方向。对于每一个天区和每一个目标星表, 估计其在 J2000.0 历元下的位置修正 ($\Delta\alpha_{2000}, \Delta\delta_{2000}$) 以及对应的自行修正 ($\Delta\mu_\alpha, \Delta\mu_\delta$)。上述修正量采用稳健统计量 (中位数) 进行估计, 以降低异常值和残余误匹配对结果的影响。

表 2. Summary of astrometric catalogs used in this work

星表	星等	星数	历元	参考系	自行	天区覆盖率	VizieR CDS ID
AC	$V \approx 6\text{--}11.0$	2,026,347	plate epoch	B1950	NO	41	I/154
AGK1	$V \approx 2.2\text{--}9.1$	5,954	mean epoch	B1875	NO	0	I/310
AGK3	$B, V \approx 6\text{--}11$	183,145	epoch of observation	B1950	YES	4	I/61B
FK4 catalogue	$V \approx 1\text{--}9.5$	1,535	B1950	B1950	YES	0	I/143
SAO	$V = 0\text{--}11$	258,997	J2000	J2000	YES	5	I/131A
ACRS	$V \approx 6\text{--}11$	320,211	J2000	J2000	YES	7	I/171
ACT	$V \approx 6\text{--}11.5$	988,758	J2000	J2000	YES	20	I/246
PPM	$V \approx 1\text{--}11.5$	378,910	J2000	J2000	YES	8	I/146, I/193
Yale	$V \approx -1\text{--}6.5$	221,338	epoch of observation	B1950	YES	5	I/141
Hipparcos	$V \approx 1\text{--}12.4$	118,218	J1991.25	J2000	YES	2	I/239
Tycho 2	$V \approx 2\text{--}12.5$	2,539,913	J2000	J2000	YES	52	I/259
USNO A2.0	$B, R = 11\text{--}20$	526,280,881	mean epoch	J2000	NO	10,707	I/252
GSC 1.2	$V \approx 6\text{--}15$	18,825,216	plate epoch	J2000	NO	383	I/254
UCAC 2	$R \approx 8\text{--}16$	48,330,571	J2000	J2000	YES	983	I/289
UCAC 4	$R \approx 7\text{--}16$	113,780,093	J2000	J2000	YES	2,315	I/322A
Gaia EDR3	$G = 3\text{--}21$		J2016	J2016	YES	0	I/350
Gaia DR2	$G = 3\text{--}21$	1,331,909,727	J2015.5	J2015.5	YES	27,098	I/345
Gaia DR1	$G = 5\text{--}20.7$	1,142,685,696	J2015	J2015	NO	23,248	I/337

NOTE—The basic properties of the astrometric catalogs included in the debiasing analysis. The sky coverage values quantify the number of sky regions with usable data in this work.

对于一条在 t 时刻给出的光学观测，其由参考星表系统误差引入的修正量表示为：

$$\begin{aligned}\Delta\alpha(t) &= \Delta\alpha_{2000} + \Delta\mu_{\alpha} \cdot (t - 2000.0) \\ \Delta\delta(t) &= \Delta\delta_{2000} + \Delta\mu_{\delta} \cdot (t - 2000.0)\end{aligned}\tag{1}$$

NOTE—其中赤经方向的修正量已包含球面几何中的 $\cos\delta$ 因子。将上述修正应用于原始观测后，历史光学观测可视为已统一至 Gaia DR3 参考框架下，从而能够在一致的误差模型中用于轨道确定与残差分析。

3.2. 修正流程与实现

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4. RESULTS AND VALIDATION

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5. DISCUSSION

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6. CONCLUSIONS

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7. DATA AND CODE AVAILABILITY

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APPENDIX

A. COMPLETE LIST OF CATALOGS AND PARAMETERS